

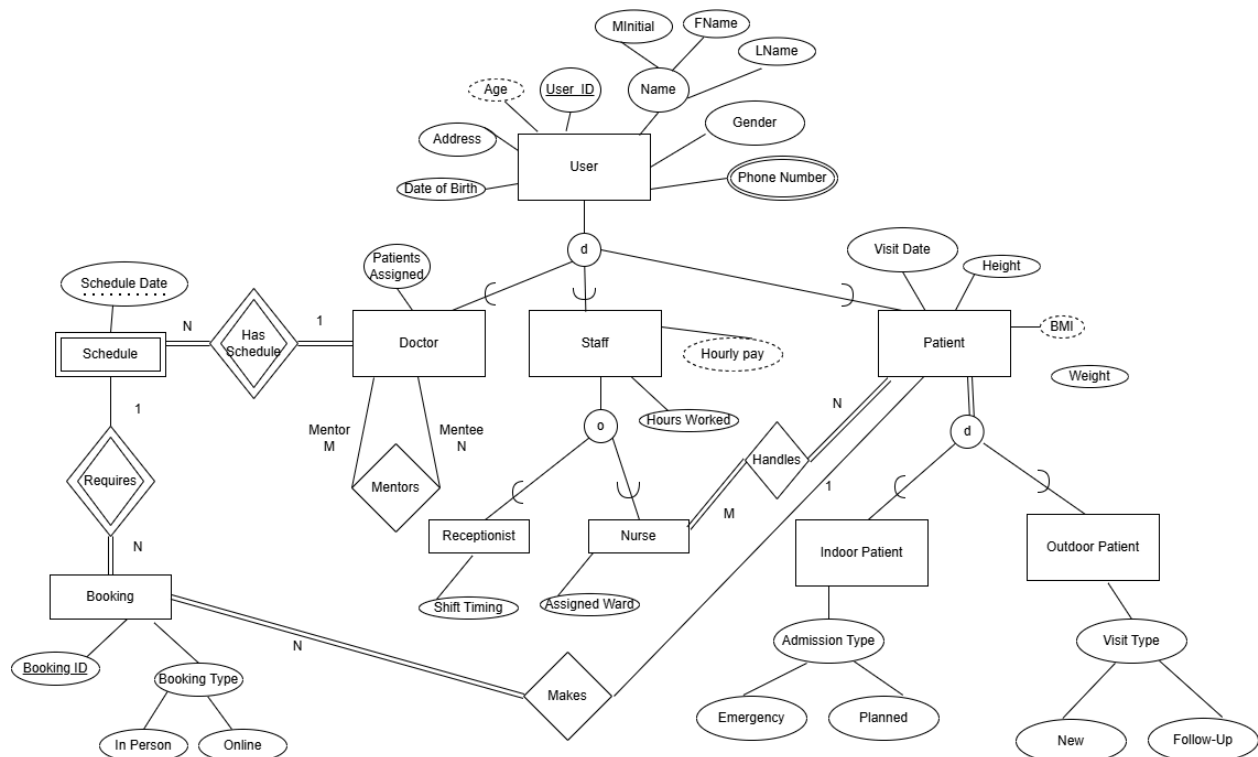
Department of Computer Science and Engineering

Course Code: CSE 370	Assessment Type: Assignment
Course Name: Database Systems	Semester: Fall 2025

Question 01 no Schema Mapping [15 Marks]

You have been hired to design a database for a Smart Healthcare Network that coordinates multiple staffs and patient admissions in the system. You need to apply appropriate rules to map the following Extended Entity Relationship (EER) diagram to a relational model schema.

Important constraint: You have to utilize at least 3 rules out of 8A,8B,8C,8D for schema mapping.



Question 02 on Normalization: [5 Marks]

Consider the relation be below:

Music_Bank (SingerID, AlbumID, SongID, SingerName, SPhone, SEmail, AName, MusicLabel, LabelAddress, RecordDate, ReleaseDate, PayAmount, SongTitle, Genre, BandID, BandName, BMgrPhone)

The primary key is underlined. The additional functional dependencies are:

FD1: SingerID $\rightarrow$ SingerName, SPhone, SEmail, BandID, BandName, BMgrPhone

FD2: AlbumID $\rightarrow$ AName, ReleaseDate, MusicLabel, LabelAddress

FD3: AlbumID, SongID $\rightarrow$ SongTitle, Genre

FD4: BandID $\rightarrow$ BandName, BMgrPhone

FD5: MusicLabel $\rightarrow$ LabelAddress

The above relation is already in 1NF. Apply normalization to normalize the relation first to 2NF and then to 3NF. Show complete schema after each normalization.

Question 03 on Normalization: [5 Marks]

Consider the relation be below:

Blood_Donation (DonorID, PatientID, RequestDate, DName, DPhone, DEmail, PName, PContact, PCondition, EmergencyLevel, DonationDate, DNextAvailability, DonorBadge, TotalDonation)

The primary key is underlined. The additional functional dependencies are:

FD1: DonorID $\rightarrow$ DName, DPhone, DEmail, DonorBadge, TotalDonation

FD2: PatientID $\rightarrow$ PName, PContact

FD3: PatientID, RequestDate $\rightarrow$ PCondition, EmergencyLevel

FD4: DonationDate $\rightarrow$ DNextAvailability

FD5: TotalDonation $\rightarrow$ DonorBadge

- Explain if the above relation in first normal form (1NF) or not? If not, apply 1NF normalization.
- Explain if the schema in (a) is in the second normal form (2NF) or not? If not, apply 2NF normalization.
- Explain if the schema in (b) is in the third normal form (3NF) or not? If not, apply 3NF normalization.

Show full schema after each normalization step.